

EXPLORE

Zingiberaceae Against Depression: The Impact of Curcuma longa and Zingiber officinale in Clinical Treatment

--Manuscript Draft--

Manuscript Number:	EXPLORE-D-26-00756
Article Type:	Review article/Systematic review/Meta-analyses
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**Cover Letter for Manuscript Submission to
*Explore: The Journal of Science & Healing***

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Date: 14th May 2026

Dear Editor,

We are pleased to submit our manuscript, titled "**Zingiberaceae Against Depression: The Impact of *Curcuma longa* and *Zingiber officinale* in Clinical Treatment**" for consideration as a review paper in the *Explore: The Journal of Science & Healing*.

The potential of ethnopharmacological interventions, particularly those derived from plants such as turmeric (*Curcuma longa*) and ginger (*Zingiber officinale*), has generated considerable interest in the treatment of depression. However, clinical consensus regarding their efficacy remains ambiguous due to significant variability across studies. Our study provides a unique, family-level comparison of species-specific effects, moving beyond isolated compound reviews to evaluate the interaction between phytotherapy and clinical health status.

Our analysis of 13 unique randomized controlled trials identified through a search of 3,167 records provides high-impact insights for the field of integrative medicine:

1. Phenotype-Dependent Efficacy: We demonstrate that *C. longa* functions as a potent "corrective adjuvant" specifically for patients with Major Depressive Disorder, but yields negligible effects in non-clinical or lifestyle-related cohorts.
2. The Ginger Severity Threshold: We resolve the long-standing statistical heterogeneity regarding *Z. officinale*. Our subgroup analysis reveals that ginger's antidepressant potential is "unlocked" only in high-severity clinical cases, whereas healthy cohorts experience a "floor effect" with no measurable change.
3. Biological Rationale: Our synthesis suggests that the antidepressant success of these plants is intrinsically linked to baseline levels of inflammation and antioxidant impairment, supporting the "Cytokine Hypothesis" of depression.
4. Technological Necessity: We highlight that high-bioavailability formulations (e.g., nano-micelles, BCM-95®) are essential moderators of clinical success, distinguishing medical-grade extracts from standard culinary powders.

As Explore serves as a leading forum for evidence-based healing and integrative medicine, we believe our findings which identify exactly which patients benefit most from specific Zingiberaceae interventions will be highly valuable to your readership of clinicians and researchers.

We confirm that this work is original, has not been published elsewhere, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission. We also confirm that we have no conflicts of interest to disclose. Additionally, the review protocol was conducted in strict accordance with PRISMA protocols and was registered with PROSPERO (ID: CRD420251053459).

Thank you for considering our manuscript for publication in Journal of Psychiatry Research. We look forward to the opportunity to contribute to the ongoing dialogue on innovative approaches to mood disorder management.

Sincerely,

A handwritten signature in black ink, appearing to read 'Unwaniah', followed by a horizontal line.

Unwaniah Abdull Rahim

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Highlights

1. Zingiberaceae species exhibit unique antidepressant effects specific to each species.
2. Turmeric/curcuminoids offer strong antidepressant benefits in cases of Major Depressive Disorder.
3. Turmeric and ginger antidepressant effects depend on baseline inflammation and oxidative stress.
4. Galangal and related family members are under-researched for depression.

Title Page

Zingiberaceae Against Depression: The Impact of *Curcuma longa* and *Zingiber officinale* in Clinical Treatment

Type of Article: Review Paper

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ABSTRACT

Background: Despite growing interest in the therapeutic potential of ethnopharmacology interventions for mental health, the efficacy of the Zingiberaceae family, specifically turmeric and ginger, remains a subject of clinical debate. This study aimed to systematically review and meta-analyse the efficacy of Zingiberaceae species in alleviating depressive symptoms. **Methods:** This review followed PRISMA guidelines. Electronic databases (PubMed, ScienceDirect, and Scopus) were searched from inception through August 2025. Randomised controlled trials (RCTs) assessing depressive symptoms were included. Data synthesis was performed using a random-effects model. **Results:** From 3,167 retrieved records, 15 RCTs met the inclusion criteria for the systematic review. 13 studies provided data for meta-analysis: nine on turmeric and four on ginger. Galangal was excluded due to a lack of studies. Result suggested that turmeric and its curcuminoids demonstrate the most reliable antidepressant effects in patients with clinically diagnosed Major Depressive Disorder (MDD) (SMD = 0.79, 95% CI: 0.44 to 1.15; $p < 0.0001$), particularly when bioavailability is enhanced. However, their effects are less consistent in cases of comorbid or milder depression. For ginger, overall potency was not significant, but it may benefit some subgroups with severe depressive symptoms (MD = 6.30, 95% CI: 0.97 to 11.63; $p = 0.02$). **Conclusion:** This study finds turmeric effective as an adjuvant for patients with MDD, whereas ginger's antidepressant effects are notable only in those with severe symptoms. More research is needed to understand their roles and to explore other Zingiberaceae plants, such as galangal, which may offer similar benefits due to shared active compounds.

Keywords: Zingiberaceae, turmeric, ginger, depression, systematic review, meta-analysis

1. INTRODUCTION

Depression is considered a mental disease characterised by disturbance of mood and affecting 3 to 5 % of the general population at one time, and 10-15% patients with depression die from suicide (Holt, 2025; Tamayo et al., 2022). Treatment should be maintained for at least 4 to 6 months after remission of symptoms to reduce the risk of relapse or recurrence (Holt, 2025). It is projected that 4.4%, or 322 million, of the total world population suffers from depression alone, making it a major cause of disease and economic burden (Chaudhari et al., 2024). By 2030, depression is predicted to become the most common disease, surpassing cancer, cardiovascular diseases, and respiratory disorders, driven by a rapid rise in diagnosed cases (Ashtree et al., 2025; Ferrari et al., 2024). Depression can be characterised by feelings of sadness, loss of pleasure, feelings of guilt, low self-esteem, and alterations in sleep patterns, loss of appetite, lack of concentration, and fatigue, which can become chronic and recurrent, ultimately leading to dysfunction in daily life activities (Cui et al., 2024). Depression is linked to

cognitive impairment and is associated with various biological mechanisms, including oxidative stress, which occurs when there is an imbalance between oxidants (such as reactive oxygen species, or ROS) and antioxidants (Aini Dania et al., 2024). This imbalance leads to inflammation, neurodegeneration, and cell damage, further affecting neurotransmitter signalling (particularly serotonin), neurogenesis, and synaptic plasticity (Lynn et al., 2025).

The rising ethnopharmacology using the Zingiberaceae family plant has been seen in promoting health with its medicinal benefits, due to its potent anti-inflammatory and antioxidant profile (Matin et al., 2025). The rationale for family plant analysis is essential because these species share common bioactive compounds, mainly phenolics and terpenoids, which are known to target the same biological pathways implicated in depression. While each species has unique secondary metabolites, the most notable members are turmeric (*Curcuma Longa* Linn), ginger (*Zingiber officinale* Roscoe), and galangal (*Alpinia galanga*), they all demonstrate potent effects with pro-inflammatory cytokines and oxidative markers (Alolga et al., 2022). Curcumin in turmeric influences neurotransmitters like serotonin and dopamine, which are essential for mood regulation (Lopresti, 2022). Ginger, rich in gingerols and shogaols, can reduce oxidative stress and inflammation linked to depression (Abdull Rahim et al., 2024). Meanwhile, galangal, noted for its flavonoids and phenols, is a powerful antioxidant that scavenges free radicals and prevent oxidative damage, improving mental health benefits (Abd Rahman et al., 2020).

With the high and growing popularity of Zingiberaceae supplements for their therapeutic effects, the findings into clinical consensus remain fragmented. Existing systematic reviews have focused primarily on the efficacy of isolated compounds, particularly curcuminoids, often overlooking other different species (Fusar-Poli et al., 2020). While *Curcuma longa* and *Zingiber officinale* are the most established in clinical practice, preliminary data on species such as *Alpinia galanga* indicate a broader therapeutic family effect that remains to be fully integrated.

Clinical trials investigating these plants yielded heterogeneous results. Therefore, this study presents a systematic review of the broader Zingiberaceae family to evaluate their therapeutic potential. While some Randomised Controlled Trials (RCTs) report significant symptom reduction, others fail to demonstrate superiority over placebo, potentially due to variations in dosage, bioavailability, and the treatment duration. Therefore, to address these discrepancies, this study presents a systematic review and meta-analysis of the effects of plants from the Zingiberaceae family on depression. This analysis aims to evaluate the collective and comparative efficacy of this botanical family rather than being restricted to a single species.

2. METHODOLOGY

2.1 Search strategy and sources

This review was developed in accordance with the PRISMA guidelines. The protocol was registered on PROSPERO (ID: CRD420251053459). The following databases were searched from inception through August 2025: PubMed, ScienceDirect, and Scopus. The following search strategy was adopted: (Curcuma OR turmeric OR Zingiber OR ginger OR Alpinia OR galanga) AND (depression OR depressive OR “mood disorder”). Only English articles were considered for this review. Grey literature or evidence not published in academic publications were excluded.

2.2 Selection procedure

All records were extracted into a web-based application known as Rayyan, a semi-automated artificial intelligence tool. Duplicates were detected and deleted. Two reviewers (UAR and MAIA), working independently and in duplicate, first screened titles and abstracts, second screening full-texts to determine eligible studies. Any doubts were resolved through consultation with a third reviewer (SNAR). The criteria based on the PICOS framework are as follows:

- 1) Population: The inclusion criteria are adult individuals in whom depressive symptoms or mood disturbances were evaluated using validated diagnostic tools or standardised depression rating scales. There are no restrictions on gender, race, or ethnicity. The exclusion criteria are child subjects and studies with restrictions on gender, race, or ethnicity.
- 2) Intervention: The inclusion criteria are Zingiberaceae family rhizome: turmeric, ginger, and galangal, as interventions, regardless of the extraction method, formulation, dosage and duration of intervention. The intervention is administered as monotherapy or adjunctive therapy along with standard antidepressant treatment, in the form of a supplement or incorporated into the diet. The exclusion criteria are interventions that use combination treatments involving herbs or plants other than those in the selected Zingiberaceae family.
- 3) Comparator: The inclusion criteria are no treatment or placebo, and conventional drugs for depressive disorder or between Zingiberaceae family plants. The exclusion criteria are other depression drugs not used in clinical settings and studies with no comparator.
- 4) Outcomes: Primary and secondary outcomes represented by psychological well-being and depressive symptoms that were evaluated by the standard measures. Outcomes measured by any study validated depression tools such as Depression Anxiety Stress Scale (DASS), General Health Questionnaire (GHQ), and Beck Depression Inventory (BDI).
- 5) Study design: Primary clinical studies with randomized controls trials and with a comparator group. The exclusion criteria are: Case-control studies, cross-sectional studies, and cohort studies are excluded. Preclinical studies (in vivo and in vitro), in silico studies, and studies

using combined preparation of the Zingiberacea family rhizome with other mentioned herbs, Studies without a separate control group. Reviews, editorial letters, book chapters, proceedings and abstracts, where the full-text is not available or cannot be obtained. Studies published in a language other than English.

2.3 Data extraction

Data extraction by UAR, MAIA, and SNAR was conducted independently, using a standardised data extraction form for the extraction variables. The following information was extracted: country, target population, sample size, age (mean/range), diagnostic criteria, main depression diagnostic tools, preparation/formulation/brand, intervention doses and duration, pre and post-treatment mean changes and standard deviations, p-value, and result of intervention vs control were extracted from the studies for meta-analytic purposes. Study authors were contacted via email to request missing data or for clarification.

2.4 Risk of Bias (RoB)

The RoB of the included studies was assessed by two independent reviewers (UAR and MAIA) using the Cochrane RoB tool (McGuinness & Higgins, 2020). Any discrepancies were resolved by the third reviewer (SNAR). The following items were considered: randomisation process, deviations from intended interventions (Blinding of participants and personnel), missing outcome data (Attrition bias), measurement of the outcome (Blinding of outcome assessment), and selective outcome reporting (Reporting bias).

2.5 Meta-analysis

Quantitative synthesis was performed using RevMan (version 5.4.1). Continuous outcomes were analysed using standardised mean difference (SMD), specifically Hedges'g, for turmeric interventions to account for the use of different depression assessment scales across studies, while mean difference (MD) was used for ginger interventions where outcome measures were comparable. To prevent analytical bias and avoid participant double-counting, the Lopresti et al. trials were combined, utilising data from the larger-sample study Lopresti et al. 2014a and Lopresti et al. 2014b for the pooled estimate. A random-effects model was applied in all analyses to account for potential clinical and methodological heterogeneity among the included studies. Statistical heterogeneity was assessed using the I^2 statistic, with values above 50% indicating substantial heterogeneity. To investigate sources of inconsistency, studies were subgrouped by health status: (a) Primary MDD, (b) Chronic inflammatory illness, and (c) Lifestyle/metabolic subtypes for curcumin; and (a) Clinical/high severity versus (b) Non-clinical/low severity for ginger. Publication bias was explored using funnel plot symmetry.

3. RESULTS

A total of 3,167 potential studies were identified from PubMed (n=274), ScienceDirect (n=2,441), and Scopus (n=452). Subsequently, 298 studies were removed due to duplication using Rayyan Software. After reviewing the titles and abstracts, 2,842 studies were excluded. The full texts of the remaining 27 studies were screened thoroughly, and 12 were further excluded because they did not meet the inclusion criteria. Finally, 15 studies were selected for inclusion in this review, comprising 10 studies on *Curcuma longa* (Daryani et al., 2025; Esmaily et al., 2015; Lopresti & Drummond, 2017; Lopresti et al., 2014a; Lopresti et al., 2014b; Lopresti et al., 2021; Mamsharifi et al., 2023; Nouri-Vaskeh et al., 2020; Saberi-Karimian et al., 2018; Sanmukhani et al., 2014), four studies on *Zingiber officinale* (Ashraf et al., 2022; Crichton et al., 2024; Crichton et al., 2023; Shabanian et al., 2018), and one study on *Alpinia galanga* (Akbarzadeh et al., 2023). A PRISMA flowchart of the article selection process is shown in Figure 1.

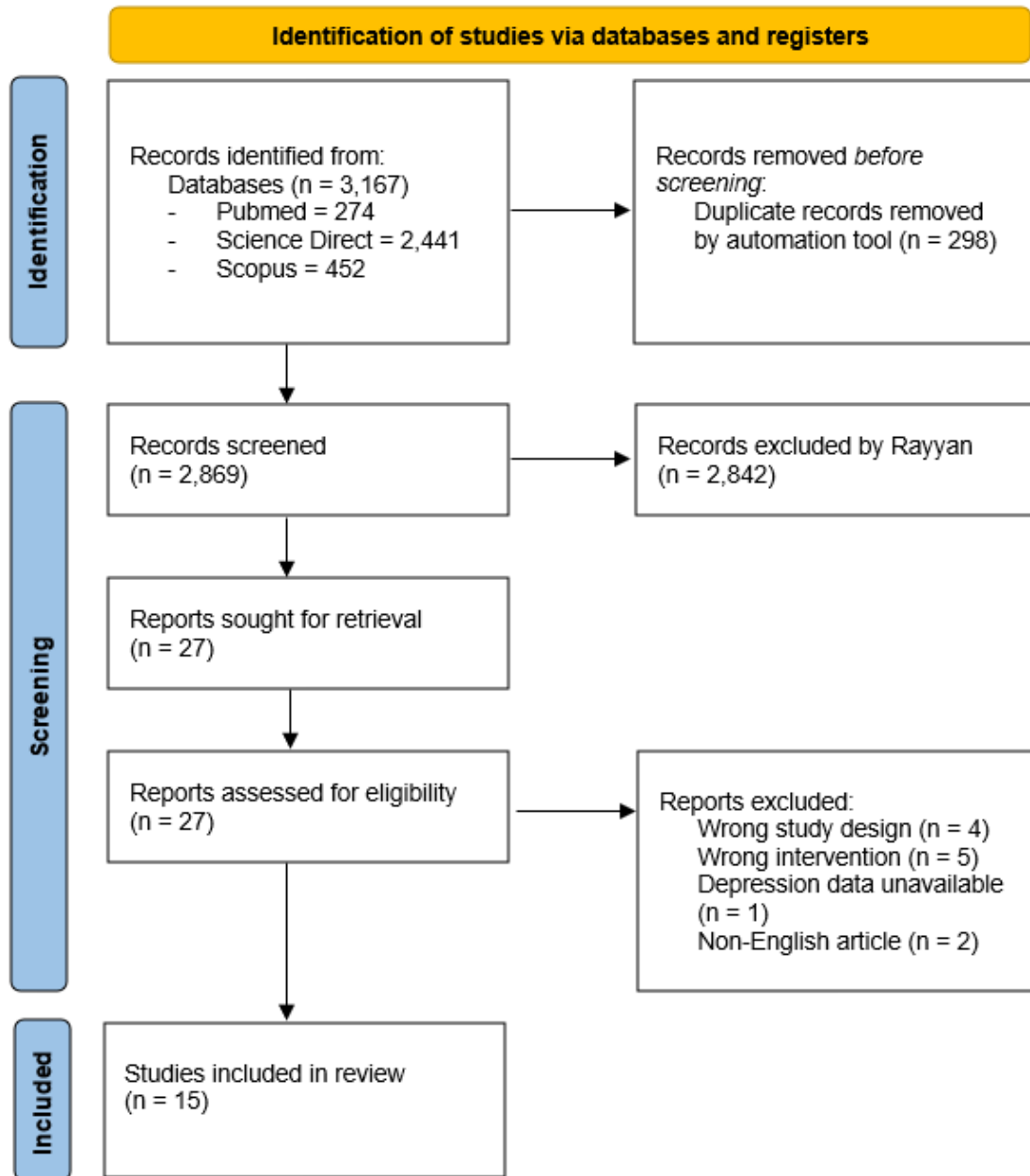


Figure 1: Selection process of the studies according to the PRISMA guideline.

Among these 15 studies that met the inclusion criteria for the systematic review, it was found that Lopresti et al. (2014b) was identified as a secondary report of the same trial (Lopresti et al, 2014a) and was excluded from the quantitative meta-analysis to avoid double-counting participants. Of these 14 trials, 13 were included in the quantitative meta-analysis (nine turmeric cohorts and four ginger cohorts). The single trial assessing *Alpinia galanga* (Akbarzadeh et al., 2023) was excluded from the meta-analysis as it was the only study found for this species, preventing statistical aggregation. Therefore, its findings was assessed qualitatively.

3.1 Study Characteristics

The characteristics of the selected studies are described in Table 1. All the included studies were randomized control trial. The studies were conducted in 3 different countries: Australia, Iran, and India. All of the clinical trial studies were conducted between 2013 and 2025, with eight studies done after 2020 (Akbarzadeh et al., 2023; Ashraf et al., 2022; Crichton et al., 2024; Crichton et al., 2023; Daryani et al., 2025; Lopresti et al., 2021; Mamsharifi et al., 2023; Nouri-Vaskeh et al., 2020). Six studies were conducted on population with depression disorder (Akbarzadeh et al., 2023; Lopresti & Drummond, 201; Lopresti et al., 2014a; Lopresti et al., 2014b; Sanmukhani et al., 2014; Shabanian et al., 2018), seven studies on populations with other clinical diagnoses (Ashraf et al., 2022; Crichton et al., 2024; Daryani et al., 2025; Esmaily et al., 2015; Nouri-Vaskeh et al., 2020; Saberi-Karimian et al., 2018), two studies on healthy individuals (Crichton et al., 2023; Lopresti et al., 2021), and the remaining population were current smokers (Mamsharifi et al., 2023). The diagnostic tools being used to measure the depression scale for the studies include ISD-SR30 (Lopresti & Drummond, 2017; Lopresti et al., 2014a; Lopresti et al., 2014b), BDI (Akbarzadeh et al., 2023; Esmaily et al., 2015; Mamsharifi et al., 2023; Saberi-Karimian et al., 2018; Shabanian et al., 2018), DASS-21 (Crichton et al., 2023; Lopresti et al., 2021), HADS (Crichton et al., 2024; Daryani et al., 2025), HAM-D17 (Sanmukhani et al., 2014), SF-36 (Nouri-Vaskeh et al., 2020), and ThySRQ (Ashraf et al., 2022).

Table 1. Characteristics of the included studies.

Study	Country	Target population (Condition)	N (Analysed)	Age (Mean / Range)	Diagnostic criteria	Main diagnostic tools	Preparation/ Formulation/ Brand	Daily dosage	Duration	Comparator	Mean Changes $\pm$ SD (Baseline to after treatment)	p-value	Intervention vs Control	Key Conclusion on Depression
<i>Curcuma longa</i> (10 studies)														
Lopresti et al. (2014a, Lopresti 2014b)	Australia	MDD	52 (27 placebo, 25 curcuminoids)	44–48	DSM-IV	IDS-SR30	BCM-95® (88% Curcuminoids)	1000 mg	8 weeks	Placebo	33.04 $\pm$ 9.39 to 22.71 $\pm$ 9.36	p = 0.043	Curcumin > Placebo	Significant changes from baseline-week 4-8. Response linked to inflammatory biomarker ET-1
Sanmukhani et al. (2013)	India	MDD	45 (14 curcumin, 16 fluoxetine, 15 combination)	33–40	DSM-IV	HAM-D ₁₇	BCM-95® (88% Curcuminoids)	1000 mg	6 weeks	Fluoxetine (20mg/day)	19.3 $\pm$ 6.50 to 13.3 $\pm$ 5.71	p = 0.77	Curcumin comparable to Fluoxetine	Curcumin is comparable to Prozac.
Lopresti et al. (2017)	Australia	MDD	111 (31 placebo, 30 high dose curcumin, 26 low dose curcumin, 24 combination)	40–46	DSM-IV, MINI 6.0	IDS-SR30	BCM-95® (88% Curcuminoids)	500–1000 mg	12 weeks	Placebo cellulose	35.29 $\pm$ 12.70 to 23.04 $\pm$ 11.69 (low dose), 34.15 $\pm$ 9.59 to 22.43 $\pm$ 11.43 (high dose)	p = 0.031	Significant antidepressant effect vs placebo.	Non-significant interaction in the IDS score between HDC and LDC
Daryani et al. (2025)	Iran	Ulcerative colitis	37 (18 curcumin, 19 placebo)	56	SCCAI	HADS	SinaCurcumin® (80 mg curcuminoid)	160 mg	4 weeks	Placebo	13.8 $\pm$ 7.2 to 11.3 $\pm$ 7.1	p = 0.06	Significant reduction vs placebo	Nano-curcumin trended toward benefit.
Nouri-Vaskeh et al. (2020)	Iran	Liver cirrhosis	58 (30 placebo, 28 curcumin)	46	MELD	SF-36 (mental health domain)	Curcumin	1000 mg	12 weeks	Placebo	57.86 $\pm$ 8.60 to 60.29 $\pm$ 7.29	p = 0.019	Significant reduction vs placebo	Improved mood in cirrhotic patients
Saberi-Karimian et al. (2018)	Iran	Metabolic Syndrome (MetS) patients	109 (37 phospholipid curcumin complex, 36 unformulated curcumin, 36 placebo)	18–65	MetS (according to IDF criteria)	BDI	Phospholipidated curcumin, unformulated curcumin	200 mg	6 weeks	Placebo	33.0 $\pm$ 13.63 to 28.75 $\pm$ 8.51 (phospholipid curcumin) 29.46 $\pm$ 10.28 to 24.22 $\pm$ 4.08 (curcumin)	p = 0.959	Non-significant change	No effect on BDI score in MetS patients

Mamsharifi et al. (2023)	Iran	Cigarette smokers	70 (35 placebo, 35 curcumin)	32–33	Fagerstrom score	BDI	Nano-Cur	80 mg	12 weeks	Placebo	24.45 ±7.06 to 24.08 ± 6.59	p = 0.31	No significant change	Nano-curcumin did not affect BDI scores in smokers.
Lopresti et al. (2021)	Australia	Digestive complaints	77 (39 placebo, 38 curcugen)	40–43	GSRS	DASS-21 (D)	Curcugen™	500 mg	8 weeks	Microcrystalline placebo	22.87 ± 16.92 to 19.28 ± 18.92	p = 0.06	No significant change in depression	Significant changes in DASS-21 score but no differences in depression.
Esmaily et al. (2015)	Iran	Obese individuals	30 (15 curcuminoids, 15 placebo)	37–38	BMI	BDI	C3 Complex® (Curcuminoids)	1000 mg	30 days	BioPerine® placebo	9.89 ± 6.50 to 9.47 ±6.02	p = 0.7	No significant changes	Curcumin did not affect BDI scores and depression index
Zingiber officinale (4 studies)														
Ashraf et al. (2022)	Iran	Primary hypothyroidism	53 (27 ginger, 26 placebo)	38–42	Controlled TSH	ThySRQ (D)	Ginger powder	500 mg	30 days	Starch placebo	1.75 ± 0.99 to 1.54 ±1.06	p = 0.376	No significant improvement	Ginger did not improve depression or low feelings
Crichton et al. (2023)	Australia	Healthy adults	51 (29 ginger, 22 placebo)	24–25	No clinical diagnosis	DASS-21	Ginger root powder (16 mg gingerols and 5 mg shogaols)	1200 mg (84 mg active compounds)	14 days	Microcrystalline cellulose placebo	1.5 ± 3.5 to 0.0 ± 3.00	p = 0.57	No significant changes	No mood effect in healthy adults.
Crichton et al. (2024)	Australia	Chemotherapy-naïve adults (Cancer)	103 (51 ginger, 52 placebo)	58–59	Receiving emetogenic chemo	HADS-D	Ginger root powder (16 mg gingerols and 5 mg shogaols)	1201 mg (84 mg active compounds)	5 days	Microcrystalline cellulose placebo	3.94 ± 2.61 to 4.97 ±3.53	p = 0.377	No significant effects	No meaningful findings on the health-related QoL and depression
Shabanian et al. (2018)	Iran	Depressed women with sexual dysfunction	131 (33 placebo, 32 ginger, 64 other plant treatments)	32–33	FSFI	BDI	Ginger oral drop (Zardband)	Not specified	8 weeks	Placebo	20.26 ± 11.39 to 13.96 ±10.52	p = 0.576	Ginger improves depression vs placebo	Ginger improved BDI scores in women.
Alpinia galanga (1 study)														
Akbarzadeh et al. (2023)	Iran	Anxiety/depressed patients with SSRI medication	60 (30 galangal, 30 placebo)	43–45	Patients on stable SSRI, IIEF	BDI	Galangal extract	500 mg	4 weeks	Avicel (starch-like) placebo	21.3 ± 5.9 to 13.0 ±4.6	p = 0.679	Both groups improved significantly	No superiority of Alpinia vs placebo.

MDD: Major Depressive Disorder, IDF: International Diabetic Federation, DSM-IV: Diagnostic and Statistical Manual-Fourth edition, HAM-D: Hamilton Depression Rating Scale, BDI: Beck Depression Inventory, DASS-21: Depression Anxiety Stress Scale- 21, HADS: Hospital Anxiety and Depression Scale, SSRI: Selective serotonin reuptake inhibitors, TSH: Thyroid stimulating hormone, ThySRQ: Thyroid Symptom Rating Questionnaire, SCCAI: Simple Clinical Colitis Activity Index, IDS-SR: Inventory of Depressive Symptomatology self-rated, GSRS: Gastrointestinal Symptom Rating Scale, MINI: Mini International Neuropsychiatric Interview, FSFI: Female Sexual Function Inventory, MELD: Model for end stage liver disease, LDSI: liver disease symptom index, SF-36: Short Form 36 Quality of Life, BMI: Body mass index, SSRI: Selective serotonin reuptake inhibitors, IIEF: International Index of Erectile Function

3.2 Effects of Turmeric on depression

There were ten clinical studies on the effect of *Curcuma longa* (or turmeric) on depression, including studies on MDD and secondary depressive symptoms in patients with other clinical diagnosis conditions. In patients diagnosed with MDD, turmeric demonstrated statistically significant superiority in anti-depressant effects over placebo in two trials using 1000mg over 8 to 12 weeks ($p=0.043$ (Lopresti et al., 2014a); $p=0.031$ (Lopresti & Drummond, 2017)). Another trial reported the same formulated curcuminoid consumption for 6 weeks was as effective as fluoxetine, but did not show a significant statistical difference between the two treatments ($p=0.77$ (Sanmukhani et al., 2014)). Similarly, another trial showed a non-significant reduction in the depression scale towards the treatment ($p>0.05$) (Lopresti et al., 2014b)).

In populations where depression was a comorbidity of physical illness, turmeric bioactive compounds showed efficacy only in patients with liver cirrhosis ($p=0.019$ (Nouri-Vaskeh et al., 2020)). In contrast, turmeric bioactive compounds were reported to have no significant anti-depressant effects in patients with metabolic syndrome (Saber-Karimian et al., 2018), obese individuals (Esmaily et al., 2015), and cigarette smokers (Mamsharifi et al., 2023). Notably, curcumin supplementation showed a positive trend towards antidepressant effects but did not achieve a statistically significant reduction in depression among individuals with digestive and irritable bowel syndrome, as well as patients with ulcerative colitis.

3.3 Effects of Ginger on depression

There were four clinical studies on the effects of antidepressant potential of *Zingiber Officinale* (or ginger), including populations with clinical depression, comorbidities as well as healthy individuals. A study by Shabanian et al reported a statistically significant reduction in BDI scores, with a mean reduction from 20.26 at baseline to 13.96 after 8 weeks of treatment (Shabanian et al., 2018). However, upon comparing the rate of improvement between groups (ginger vs placebo), the difference was not statistically significant ($p=0.576$), suggesting ginger's effectiveness over placebo effect in this population requires further validation.

In populations where mood symptoms were secondary to medical conditions, the efficacy of ginger bioactive compounds was limited. Two trials reported no significant improvement in depression or mood after treatment with ginger powder supplements compared to the placebo group in patients with hypothyroidism ($p=0.3376$) (Ashraf et al., 2022) and cancer patients ($p=0.377$) (Crichton et al., 2024). The effects of ginger on healthy individuals similarly did not significantly alter mood or depression scores on DASS-21, which may be attributed to the baseline depression scores being already within normal range ($p=0.57$) (Crichton et al., 2023).

3. 4 Effects of Galangal on depression

The effects of *Alpinia galanga* (or galangal) were evaluated in a cohort of depressed/anxiety male patients with SSRI-induced sexual dysfunction (Akbarzadeh et al., 2023). Although galangal and placebo groups' depression scale improved significantly (mean changes from baseline 21.3 to 13.0 after treatment), the difference in the rate of improvement between the galangal and placebo groups was not statistically significant ($p=0.679$). This suggests that while galangal is well tolerated, it did not improve depression in patients already managed with SSRI medications.

3.5 Risk of Bias

The risk of assessment for all 15 included studies is presented in Figure 2. Clinical trials included in this review are presented in Figure 2. The domains for the randomisation process (D1), outcome measurement (D4), and selection of the reported result (D5) were rated as low risk of bias for all studies. Deviations from the intended intervention (D2) were rated low in all studies, apart from Sanmukhani et al. (2014), which was limited by an observer-masked study design, whereas participants were not blinded to the intervention, potentially introducing performance bias due to participant expectations. Missing outcome data was assessed high in one study (Nouri-Vaskeh et al., 2020), where approximately 17% of the patients who enrolled in the trial were excluded from the final analysis due to lost to follow-up and mortality, while the rest of the studies were low risk. The authors utilised a complete-case analysis rather than an intention-to-treat (ITT) approach, and did not perform imputation for missing data. Since Lopresti et al. (2014a) and Lopresti et al. (2014b) originate from the same randomised controlled trial, their risk of bias was assessed together, utilising the primary efficacy report and the secondary biomarker analysis to provide a comprehensive evaluation of the study's quality.

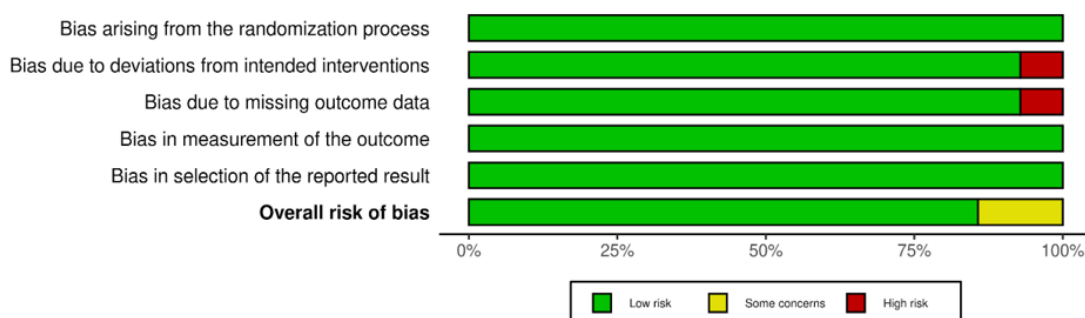
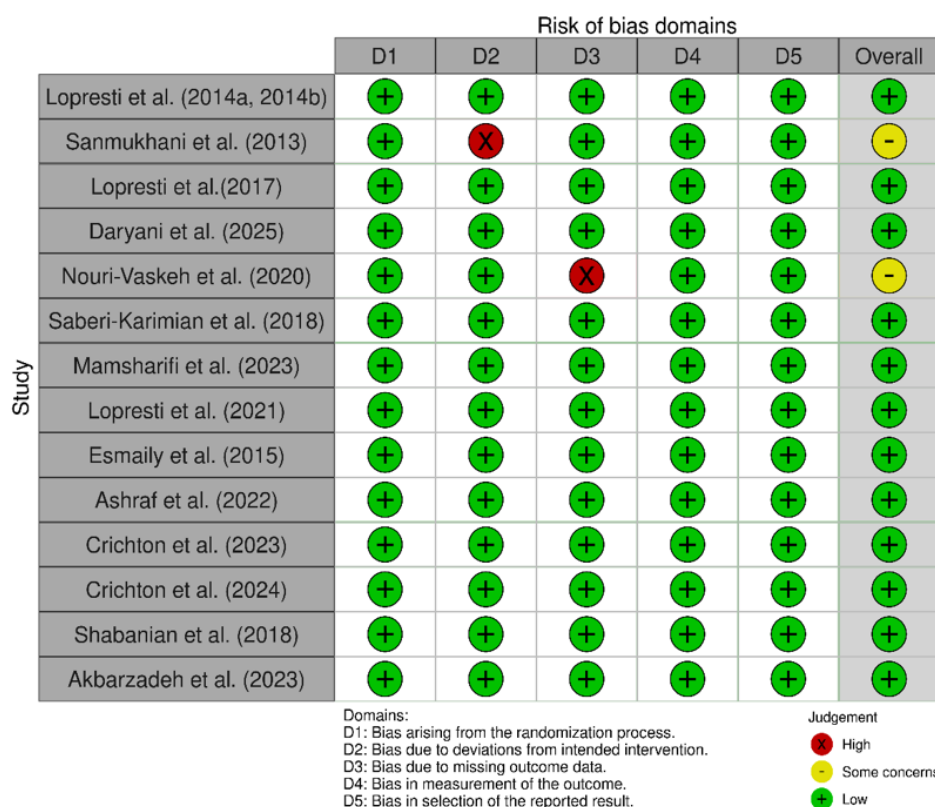


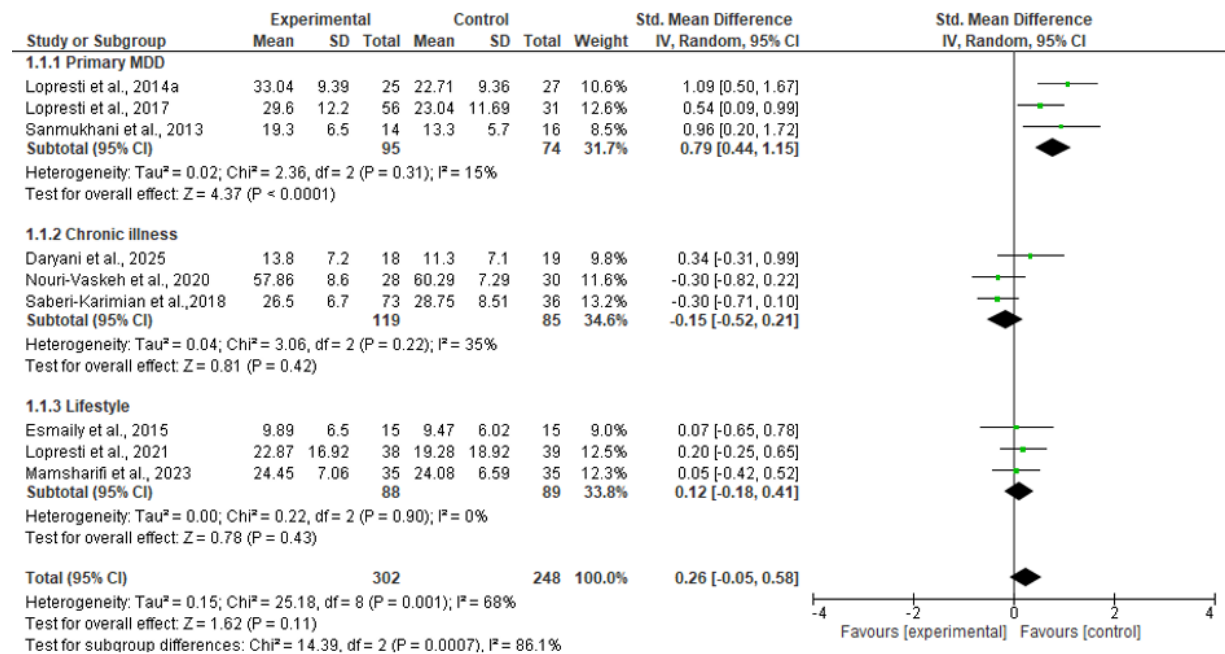
Figure 2: Risk of Bias (RoB) assessment of included studies.

3.6 Meta-Analysis

A total of 13 randomised controlled trials were included in the meta-analysis, comprising 9 studies evaluating turmeric and 4 studies evaluating ginger (Figure 3). For turmeric, the overall pooled analysis demonstrated a non-significant effect on depressive symptoms (SMD = 0.26, 95% CI: -0.05 to 0.58; $p = 0.11$), with moderate heterogeneity ($I^2 = 68\%$). Subgroup analysis revealed a statistically significant effect in the primary MDD subgroup (SMD = 0.79, 95% CI: 0.44 to 1.15; $p < 0.0001$) with low heterogeneity ($I^2 = 15\%$). The chronic illness subgroup showed no significant effect (SMD = -0.15, 95% CI: -0.52 to 0.21; $p = 0.42$; $I^2 = 35\%$), and the lifestyle subgroup demonstrated no significant effect (SMD = 0.12, 95% CI: -0.18 to 0.41; $p = 0.43$; $I^2 = 0\%$). The test for subgroup differences was statistically significant ($p = 0.0007$, $I^2 = 86.1\%$). For ginger, the overall pooled analysis showed no statistically significant effect (MD = 0.49, 95% CI: -0.90 to 1.88; $p = 0.49$), with moderate

heterogeneity ($I^2 = 73\%$). However, subgroup analysis demonstrated a significant effect in the clinical/high severity subgroup (MD = 6.30, 95% CI: 0.97 to 11.63; $p = 0.02$), whereas the non-clinical/low severity subgroup showed no significant effect (MD = 0.10, 95% CI: -1.01 to 1.20; $p = 0.87$; $I^2 = 66\%$). Subgroup differences were statistically significant ($p = 0.03$, $I^2 = 79.9\%$). Funnel plot analysis did not demonstrate clear asymmetry; however, interpretation is limited due to the relatively small number of included studies (Figure 4).

(a)



(b)

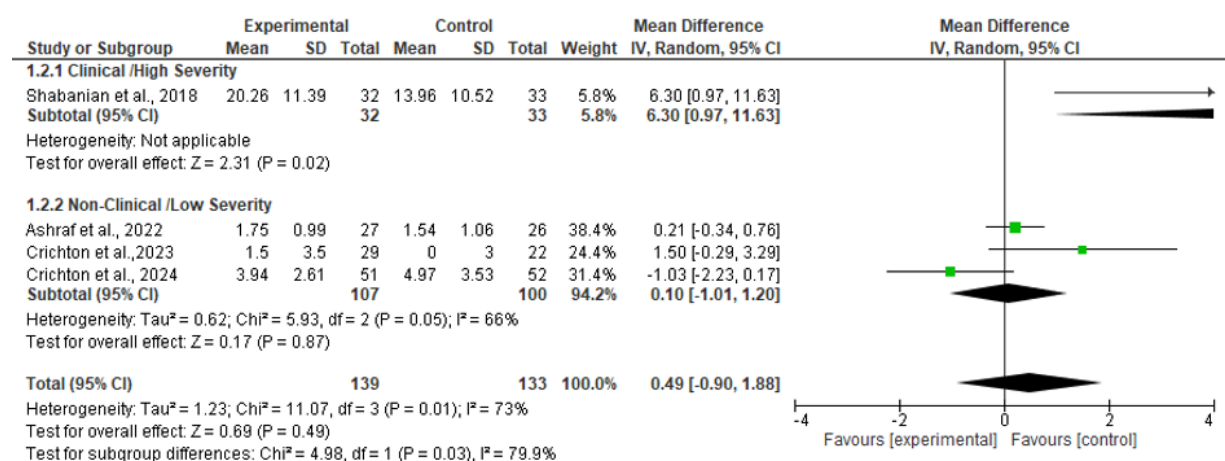


Figure 3: Forest plots of effects of Zingiberaceae family plants on depression symptoms. Results from two meta-analyses. (a) Turmeric and (b) Ginger.

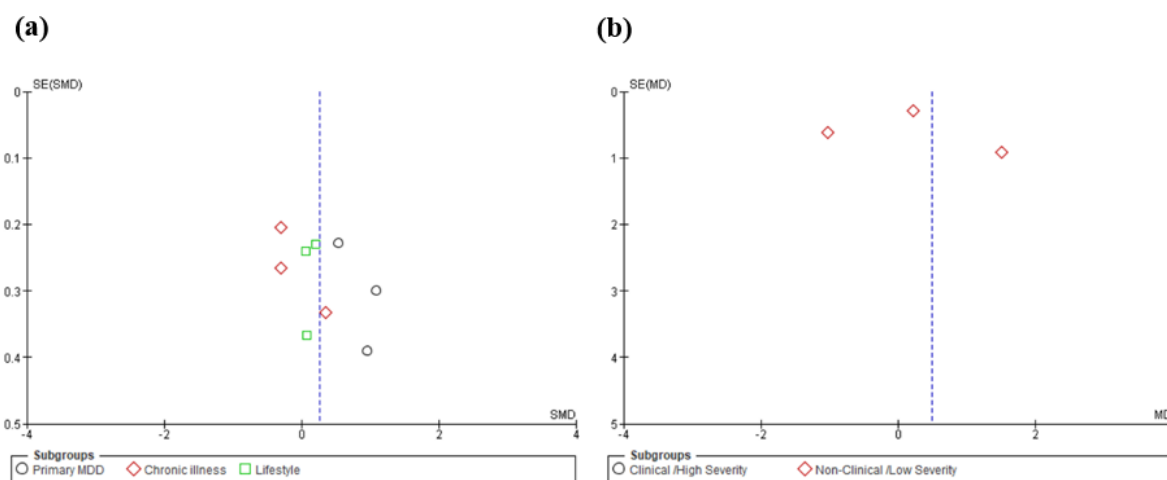


Figure 4: Funnel plots of effects of Zingiberaceae family plants on depression symptoms. Results from two meta-analyses. **(a)** Turmeric and **(b)** Ginger.

4. DISCUSSION

This systematic review and meta-analysis of the included studies provides an evaluation of the Zingiberaceae family plants, including turmeric, ginger and galangal, in the management of depression symptoms. Our findings suggest that turmeric and ginger, along with their active compounds, are the two most culturally and clinically integrated plants of this family. Moreover, while active compounds from both plants have proven their antidepressant properties, their effectiveness is not consistent across all cases. Instead, their effectiveness is highly modulated by the clinical status of the population and the disease severity.

Turmeric contains the bioactive curcuminoid complex, primarily made up of curcumin, dimethoxycurcumin, and bisdemethoxycurcumin, with concentrations of 62-90, 9-23, and 0.3-14 mg/g of turmeric, respectively (El-Saadony et al., 2022). Curcumin is found to be the most potent among the curcuminoid complex (Abdull Rahim et al., 2024). Curcuminoids are recognised in numerous studies as potent in modulating neurotransmitter levels, inflammatory pathways, excitotoxicity, neuroplasticity, hypothalamic-pituitary-adrenal disturbance, and oxidative and nitrosative stress, all of which can be involved in MDD pathophysiology (Ramaholimihaso et al., 2020). Osimo et al. (2019) found that about 25% of patients with depression have CRP levels above 3 mg/L, which is considered a marker for low-grade inflammation. By reducing inflammation, turmeric/curcuminoids could help alleviate inflammatory symptoms in individuals with depression (Spanoudaki et al., 2024). In this review, curcuminoids emerged as the most consistently studied supplements as interventions in clinical trials with a statistically significant improvement in patients with depressive symptoms. The differences in bioavailability and formulation are central to this variable. Trials using formulated products, such as BCM-95® (with 88% curcuminoids), often report more consistent pharmacokinetic profiles and

sometimes stronger biological effects, reinforcing the idea that formulation influences differences between studies (Lopresti & Drummond, 2017; Lopresti et al., 2014a; Lopresti et al., 2014b; Sanmukhani et al., 2014). Meanwhile, the subgroup analysis showed that the treatment was most effective in patients with a primary diagnosis of MDD, with very low heterogeneity. However, in pooled analyses across trials for liver cirrhosis, metabolic syndrome, colitis, and cancer, where depression is present as secondary to comorbid conditions, turmeric bioactive compounds showed inconsistent and non-significant antidepressant effects (Daryani et al., 2025; Nouri-Vaskeh et al., 2020; Saberi-Karimian et al., 2018). Similar results were found among trials with lifestyle-related conditions, such as smokers, obese individuals, and those with irritable bowel syndrome (Esmaily et al., 2015; Lopresti et al., 2021; Mamsharifi et al., 2023). As reported by Lopresti et al. (2014b), curcuminoids have the ability to influence Endothelin-1 and Leptin, suggesting a vascular and metabolic component to their mood-altering effects. Another key mechanism is the anti-inflammatory effect of curcuminoid through inhibiting the NF- κ B pathway. This compound reduces the production of pro-inflammatory cytokines, including interleukin-6 and interleukin-1 β . These markers are frequently elevated in patients with major depressive disorder and are known to drive inflammatory-mediated mood changes (Fusar-Poli et al., 2020). This aligns with biological factors, indicating that systemic comorbidities and disease-specific pathophysiology influence mood response to anti-inflammatory nutraceuticals, possibly due to differences in absorption, metabolism, or inflammatory factors (Mechlińska et al., 2023).

Ginger and its main constituents, gingerols and shogaols, demonstrated pleiotropic effects in anti-oxidative and anti-inflammatory effects (Bischoff-Kont & Fürst, 2021). Ginger acts as an antidepressant due to its potent antioxidant properties and its ability to enhance serotonin production (Lynn et al., 2025). The antioxidant activity works by donating electrons or hydrogen atoms to free radicals, thereby inhibiting the formation and activity of reactive oxygen species (ROS), hence improving cognitive impairment in depression (Lynn et al., 2025). Mendóza et al. (2026) reported that the active compounds in ginger possess neuroprotective effects, which explain their anxiolytic and antidepressant effects. Nevertheless, the clinical evidence for ginger in this review remains highly variable, with many trials involving small sample sizes. The clinical effects observed are inconsistent and do not show significant differences from placebo overall. However, ginger's effectiveness became evident when the severity of clinical depression severity was analyzed in subgroups. The subgroup analysis suggests that the initial severity of symptoms might affect how patients respond to treatment, though these findings should be interpreted cautiously. In the group of depressed women who had moderate to severe clinical depression, ginger worked well, leading to a drop in their BDI scores (Shabanian et al., 2018). But in the trials involving healthy adults, hypothyroid patients, or chemotherapy-naïve individuals, with milder depressed symptoms, ginger appeared to have no effect (Ashraf et al., 2022; Crichton et al., 2024; Crichton et al., 2023).

1 However, several factors may influence ginger's effects on depression in these trials. Razak et al. (2023)
2 reported that ginger bioactive compounds influence neurotrophic and inflammatory pathways related
3 to mood disorders, as ginger's effects on the CNS are significant when these signalling pathways are
4 impaired. This suggests that ginger might be more effective as a corrective adjuvant for clinical
5 neurobiological dysregulation rather than serving as a general mood booster for people without
6 depression. Furthermore, the extended duration of the clinical trial (8 weeks) compared to shorter trials
7 indicates that a longer treatment period may be needed for ginger to demonstrate a measurable
8 antidepressant effect. Additionally, variations in bioavailability, extraction methods, and dosing could
9 also contribute to the inconsistent effects observed with ginger. Moorkoth et al. (2021) reported that
10 ginger provides short-term benefits and requires a long period to influence antidepressant effects by
11 increasing brain monoamines in animal models. In a trial reported by Anh et al. (2020) following a
12 three-month intervention, the ginger treatment group showed a significantly greater effect than the
13 placebo group. Given the small number of studies and the variability in extraction methods and dosing,
14 these population-specific benefits remain speculative and require confirmation in larger, harmonised
15 clinical trials. However, as this clinical subgroup is based on limited data, this conclusion remains
16 tentative.

17
18 The results of the meta-analysis create a clear evidence-based between turmeric and ginger for the
19 management of depressive symptoms. Turmeric and its bioactive compounds did not show a significant
20 overall pooled effect; however, subgroup analysis revealed a strong and clinically meaningful
21 antidepressant effect specifically in patients with primary MDD. This finding highlights the importance
22 of subgroup stratification in uncovering clinically relevant effects that may be masked in overall pooled
23 analyses. The antidepressant effects of turmeric active compounds, including curcumin, may be
24 attributed to their anti-inflammatory and neuroprotective mechanisms, such as modulation of the
25 hypothalamic–pituitary–adrenal (HPA) axis, enhancement of brain-derived neurotrophic factor
26 (BDNF), and inhibition of the NLRP3 inflammasome. The absence of effect in chronic illness and
27 lifestyle-related subgroups suggests that underlying disease mechanisms significantly influence
28 treatment response. In contrast, ginger did not demonstrate a consistent antidepressant effect in the
29 overall analysis. However, subgroup findings indicate that ginger may exert significant therapeutic
30 benefits in populations with clinically severe depression, while remaining ineffective in non-clinical
31 cases. This suggests that baseline severity plays a key role in treatment responsiveness. The observed
32 heterogeneity in ginger studies is likely driven by differences in study populations, intervention
33 protocols, and formulation variability. Additionally, pharmacokinetic limitations such as variable
34 absorption and bioavailability may contribute to inconsistent central nervous system effects despite
35 known anti-inflammatory properties. However, several limitations should be considered. The number
36 of studies within some subgroups was limited, and moderate heterogeneity remained in certain analyses.

1 Additionally, variability in formulations and outcome measures may have influenced the pooled
2 estimates.

3
4 Aside from turmeric and ginger, this review also reported on another member of the Zingiberaceae
5 family, Galangal (*Alpinia galanga*), which is less commonly studied for depression. Galangal contains
6 high levels of 1'S-1'-acetoxychavicol acetate (ACA), which exhibits neuroprotective and antioxidant
7 effects. It also helps improve CNS disorders, enhance mental health, and alleviate mood issues (Zhang
8 et al., 2021). In this review, galangal was tested in a highly specific cohort of men suffering from
9 depression and anxiety who were already using SSRIs (Akbarzadeh et al., 2023). While the trials
10 reported reductions in BDI scores, the lack of statistical significance between groups may indicate
11 a high placebo response. From a clinical standpoint, this suggests that galangal is safe as an adjuvant,
12 but more human trials are still required to ascertain its true benefits for depression. The limited
13 data represent a significant research gap. These plants share many of the bioactive profiles of turmeric
14 (curcuminoids) and ginger (gingerols) while containing unique sesquiterpenes that may offer distinct
15 advantages.

16 17 **5. CONCLUSION**

18
19 In conclusion, our results provide compelling evidence for the efficacy of the Zingiberaceae family in
20 reducing the severity of depressive symptoms. Turmeric and its curcuminoids demonstrate the most
21 reliable antidepressant effects in patients with clinically diagnosed MDD, particularly when
22 bioavailability is enhanced. However, their effects are less consistent in cases of comorbid or milder
23 depression. Ginger generally does not generate a strong overall potency but may benefit specific
24 subgroups with severe depressive symptoms. Conversely, galangal has preliminary, limited evidence
25 highlighting the necessity for more comprehensive research.

26
27 Ultimately, the Zingiberaceae family provides a strong, low-risk option for incorporation into adjuvant
28 antidepressant treatments. While turmeric and ginger should not entirely replace standard
29 pharmacotherapy, they provide a statistically significant enhancement in clinical effectiveness for
30 patients with major depressive disorder. Their role is best defined as a biological support system that is
31 most effective at targeting the specific inflammatory and neuroendocrine imbalances present in
32 moderate-to-severe clinical cases.

33 34 **Data availability**

35 The data that support the findings of this study are available on request from the corresponding
36 author.

Acknowledgements

None

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